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Estimates of (convective core) masses, radii, and relative ages for ~14,000 *Gaia*-discovered gravity-mode pulsators monitored by TESS

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ABSTRACT

Context. Gravito-inertial asteroseismology saw its birth from the 4-year-long light curves of rotating main-sequence stars assembled by the *Kepler* space telescope. High-precision measurements of internal rotation and mixing are available for about 600 stars of intermediate mass so far that are used to challenge the state-of-the-art stellar structure and evolution models.
Aims. Our aim is to prepare for future large ensemble modelling of gravity-mode pulsators by relying on a new sample of such stars recently discovered from the third Data Release of the *Gaia* space mission and confirmed by space photometry from the TESS mission. This sample of potential asteroseismic targets is about 23 times larger than the *Kepler* sample.
Methods. We use the effective temperature and luminosity inferred from *Gaia* to deduce evolutionary masses, convective core masses, radii, and ages for ~14,000 gravity-mode pulsators classified as such from their nominal TESS light curves. We do so by constructing two dedicated grids of evolutionary models for rotating stars with input physics from the asteroseismic calibrations of *Kepler* γ Dor pulsators. These two grids consider the distribution of initial rotation velocities at the zero-age main sequence deduced from gravito-inertial asteroseismology, for two extreme values found for the metallicity of γ Dor stars deduced from spectroscopy ([M/H] = 0.0 and −0.5).
Results. We find the new gravity-mode pulsators to cover an extended observational instability region covering masses from about 1.3 $M_\odot$ to about 9 $M_\odot$. We provide their mass-luminosity and mass-radius relations, as well as convective core masses. Our results suggest that oscillations excited by the opacity mechanism occur uninterruptedly for the mass range above about 2 $M_\odot$, where stars have a radiative envelope aside from thin convection zones in their excitation layers.
Conclusions. Our evolutionary parameters for the sample of *Gaia*-discovered gravity-mode pulsators with confirmed modes by TESS offer a fruitful starting point for future TESS ensemble asteroseismology once a sufficient number of modes is identified in terms of the geometrical wave numbers and overtone for each of the pulsators.

Key words. Asteroseismology – Stars: oscillations (including pulsations) – Stars: interiors – Stars: evolution – Methods: numerical – Catalogs

1. Introduction

Data Release 3 (DR3, Gaia Collaboration et al. 2023b) of the ESA *Gaia* space mission (Gaia Collaboration et al. 2016b,a, 2018) contains the time-series photometry of millions of variable stars classified and delivered by Coordination Unit 7 (CU 7, Eyer et al. 2023). Even if the mission was not specifically designed for it, *Gaia*'s sparsely sampled DR3 photometric light curves revealed excellent capacity to discover a multitude of new non-radial pulsators. In particular, Gaia Collaboration et al. (2023a, hereafter called Paper I) discovered more than 100,000 of such pulsators occurring along the main sequence, where a variety of excitation mechanisms are known to be active (Aerts et al. 2010, Chapter 2).

In a follow-up study to Paper I, Aerts et al. (2023, hereafter Paper II) considered the more than 15,000 newly discovered *Gaia* DR3 candidate gravity (g-)mode pulsators. These turned out to have a dominant amplitude above about 4 mmag, which was found to mark the detection threshold for nonradial oscillations with significant frequencies occurring in the sparse DR3 light curves. The CU 7 algorithms assigned these pulsators to either the class of Slowly Pulsating B-type stars (SPB stars hereafter, Waelkens 1991; Waelkens et al. 1998; Aerts et al. 1999; De Cat & Aerts 2002) or the class of γ Doradus stars (γ Dor stars hereafter, Kaye et al. 1999; Handler 1999; Uytterhoeven et al. 2011). Both these classes consist of multiperiodic low-frequency (typically between 0.5 d^{-1} and 3 d^{-1}) g-mode pulsators with internal rotation frequencies covering from slow to very fast compared to the frequencies of the excited modes (Pápics et al. 2017; Aerts et al. 2019; Li et al. 2020).

Since the era of high-cadence space photometry started, thousands of pulsators have been discovered from their high-precision (down to μmag) uninterrupted space photometric light curves. These are mainly assembled with the NASA *Kepler* and Transiting Exoplanet Survey Satellite (TESS) space telescopes (see Kurtz 2022, for an extensive review). Even though thousands of g-mode pulsators were discovered from *Kepler* data (Blomme et al. 2010; Debosscher et al. 2011; Uytterhoeven et al. 2011), only about 700 were characterised asteroseismically so far (Li et al. 2020; Pedersen et al. 2021, and Aerts 2021 for a summary), because this requires proper identification of the mode geometry of several g modes per star. While frequency analysis methods from high-cadence light curves are well established (Aerts et al. 2010, Chapter 5) and easy to apply (e.g. Bowman 2017; Van Beeck et al. 2021), identification of the corresponding mode geometry for each of the significant frequencies is not a straightforward task. For g modes, such identification can be established from the detection of so-called period spacing patterns revealed by modes of consecutive radial overtone, as highlighted in the theory papers by Miglio et al. (2008); Bouabid et al. (2013) and Van Reeth et al. (2015a). This potential was first put into practice for two SPB stars monitored during five months by the CoRoT mission (Degroote et al. 2010; Pápics et al. 2012).

The real breakthrough in g-mode asteroseismology of main-sequence stars came from the 4-year-long *Kepler* light curves. These data opened up the new sub-field of gravito-inertial asteroseismology (e.g. Pápics et al. 2014; Van Reeth et al. 2015a,b; Moravveji et al. 2015, 2016; Schmid & Aerts 2016; Van Reeth et al. 2016; Ouazzani et al. 2017; Van Reeth et al. 2018; Szewczuk & Daszyńska-Daszkiewicz 2018; Wu et al. 2018; Wu & Li 2019; Mombarg et al. 2019; Ouazzani et al. 2019; Li et al. 2020; Sekaran et al. 2021; Mombarg et al. 2021; Pedersen et al. 2021; Szewczuk et al. 2021, 2022; Pedersen 2022b,a; Fritzewski et al. 2024b,a; Li et al. 2024). This led to the major conclusion that single intermediate-mass stars have nearly rigid rotation throughout the long core-hydrogen burning phase of their evolution, with levels of differentiality between the near-core region and the stellar surface typically below 10 per cent (Li et al. 2020). The same conclusion holds for rotation frequencies of the core (Saio et al. 2021). These findings required revisions of the stellar evolution theory of such stars in regards to the transport of angular momentum (Aerts et al. 2019; Fuller et al. 2019; Moyano et al. 2024).

The excitation of g modes in SPB stars is linked to the κ-mechanism, operating around the iron opacity peak in the partial ionisation zone of heavy elements such as iron and nickel at a temperature of about 200 kK. This mechanism drives an efficient heat-engine cycle, exciting slow eigenmodes with periods of about one day to a couple of days. In γ Dor stars, the dominant excitation mechanism is believed to be convective flux blocking. This mechanism operates at the interface between the radiative zone and the thin convective outer envelope, situated in an area with a temperature from about 200 kK to 500 kK (Guzik et al. 2000; Dupret et al. 2005). However, convective flux blocking alone cannot explain the observed hotter γ Dor stars, suggesting that the κ-mechanism also plays a role in these stars (Xiong et al. 2016). The current commonly used instability regions for the SPB stars (Szewczuk & Daszyńska-Daszkiewicz 2017) and γ Dor stars (Dupret et al. 2005) suggest a gap between the red edge of the SPB instability region and the blue edge of the γ Dor strip, even when considering time-dependent convection and adopting a range of mixing-length parameters. Yet, g-mode pulsators are observed to fall within this gap (Balona & Ozuyar 2020; Balona 2024, and Paper I) indicating that at least some of the astrophysical properties of these stars are still not fully captured by the models and/or the theory of mode excitation.

In this paper, we estimate masses, convective core masses, radii, and ages (in terms of the central hydrogen-mass fraction) of the largest sample of g-mode pulsators so far. We investigate their properties from the dominant oscillation mode discovered in the *Gaia* DR3 data and meanwhile confirmed by the TESS space photometry (Hey & Aerts 2024). Combining the information in these space photometric light curves allows us to map the instability regions more carefully than has been done so far and to decide whether there is indeed a lack of stars with excited g modes in the mass regime between roughly 2 and 3 $M_\odot$. We discuss the main astrophysical properties of the ~14,000 *Gaia* DR3 g-mode pulsators confirmed and reclassified as such from TESS photometry. We also provide their potential for future asteroseismic forward and inverse modelling.

Section 2 treats the sample selection while Sect. 3 focuses on our modelling approach, which is based on two grids of stellar models, one for solar metallicity and one for a metallicity of [M/H] = −0.5. Section 4 covers the derived parameters, including the mass, convective core mass, radius, and age. We end the paper with our conclusions in Sect. 5.

2. Sample selection

It was shown in Paper II that the *Gaia* DR3 candidate g-mode pulsators have similar properties as the genuine SPB or γ Dor pulsators observed with the *Kepler* space telescope, which delivered 4-year-long high-cadence μmag-precision light curves (Gilliland et al. 2010; Kurtz 2022). This resemblance not only concerned their fundamental parameters, but also the dominant frequency and its amplitude. Nevertheless, these similarities are not a solid proof that the candidate pulsators identified by Paper I are genuine class members, because the *Gaia* DR3 data only delivered one or two significant frequencies. Moreover, the Fourier Transforms of the *Gaia* DR3 light curves reveal instrumental frequencies at mmag level, particularly for frequencies above the satellite spinning frequency of about 4 d^{-1}. Regardless whether most of the gravito-inertial modes in SPB and γ Dor stars occur below this satellite frequency, it is difficult to exclude that the secondary frequencies in the *Gaia* DR3 light curves deduced in Paper I originate from the satellite instead of from the target stars.

In order to confirm (or refute) the nature of the *Gaia* DR3 candidate pulsators, Hey & Aerts (2024) considered high-cadence light curves assembled by the nominal TESS mission (having a time base between 27 d and 352 d). They relied on all the candidate pulsators found in Paper I having data available in the homogeneously reduced TESS–*Gaia* light curve (TGLC) catalogue produced from full-frame images by Han & Brandt (2023). A total of 58,970 stars out of the 106,207 *Gaia* DR3 candidate pulsators classified in Paper I have TGLC light curves. Analyses of this data revealed that more than 70 per cent of the 58,970 stars have the same dominant frequency in the completely independent TGLC and *Gaia* DR3 light curves. Moreover, Hey & Aerts (2024) reported that almost all these stars turn out to be genuine multiperiodic pulsators.

A reclassification of these 58,970 TESS-confirmed pulsators was performed by Hey & Aerts (2024). Here, we take all stars they classified as γ Dor/SPB or hybrid pulsator with more than two observed significant independent frequencies in the TESS data. This selection criterion is based on earlier classification work for highly sampled photometric light curves assembled from space. Debosscher et al. (2009) and Blomme et al. (2010) showed that a classifier based on the demand of having at least three independent frequencies is very effective in selecting γ Dor

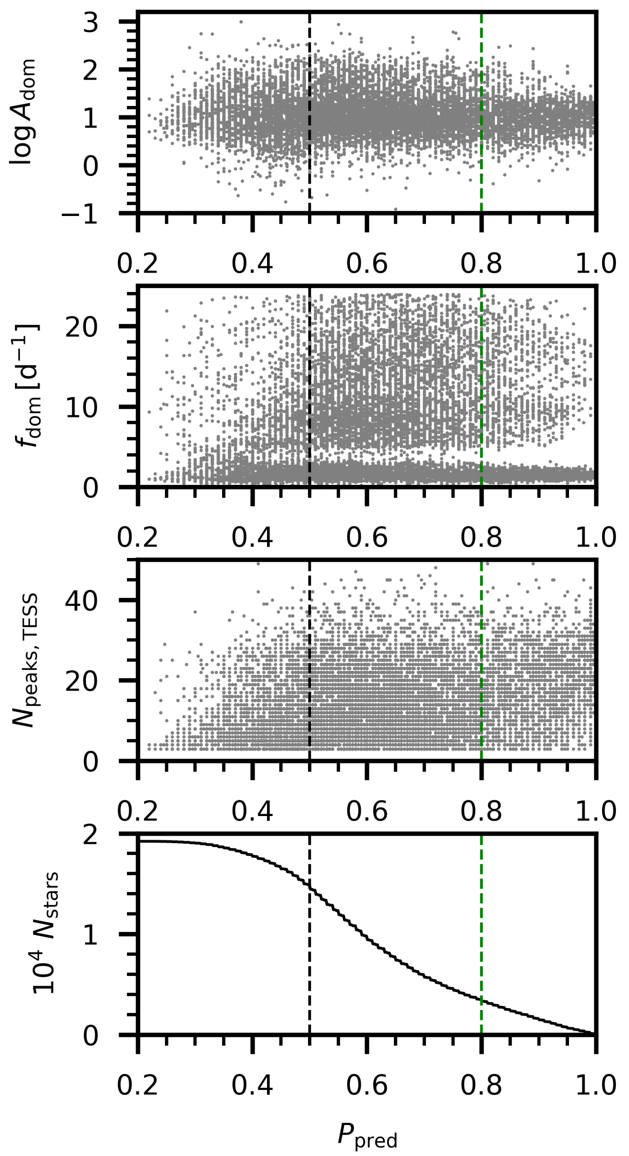

Fig. 1: Pulsational characteristics of the sample. From top to bottom: amplitude of dominant frequency in the TESS light curve (in mmag), dominant frequency, and number of significant frequencies in the nominal TESS data. The bottom panel shows the number of stars above a given threshold on the classification probability $P_{\rm pred}$ from Hey & Aerts (2024). Vertical dashed lines indicate the used threshold in $P_{\rm pred}$ for the different samples.

pulsators from space photometry (Tkachenko et al. 2013; Van Reeth et al. 2015b).

For our study, we wish to maximise the sample size of *Gaia* g-mode pulsators. However, we do not want to include too many contaminants according to the confusion matrix of the random forest classifier used by Hey & Aerts (2024). Aside from demanding three significant frequencies, which greatly helps to avoid eclipsing binaries or rotational variables, we inspect the properties of the dominant frequency commonly present in the *Gaia* and TESS photometry. Figure 1 shows this frequency, its amplitude, and also the total number of significant frequencies in the nominal TESS light curve as a function of the classification probability from Hey & Aerts (2024). We achieved maximal ranges of frequencies and amplitudes at $P_{\rm pred} \simeq 0.5$. For this reason, we worked with the sample of stars having $P_{\rm pred} \geq 0.5$.

Furthermore, we eliminated pulsators whose effective temperature locates them outside the two grids of stellar models discussed in Sect. 3. This resulted in a total of ∼14,000 new confirmed *Gaia* DR3 g-mode pulsators, 13,682 stars when we assume a metallicity of [M/H] = −0.5, and 14,072 stars when we assume [M/H] = 0, to be exact. We also investigated the effect of using a lower threshold $P_{\rm pred} \geq 0.2$ and a higher threshold $P_{\rm pred} \geq 0.8$, on the probability of the star being a g-mode or hybrid pulsator, which yielded a total of 18,132 (18,634) and 3,165 (3,195) stars, respectively for [M/H] = −0.5 (0.0). These numbers include cuts that we impose to exclude stars that fall outside of the used grids of stellar models, as we will discuss later.

The ∼14,000 stars with $P_{\rm pred} \geq 0.5$ of being a g-mode pulsator cover one global observational instability region as shown in Fig. 2. This observational 'combined *Gaia*-DR3/TESS' g-mode instability strip is far more extended than the joint coverage of various theoretically predicted individual strips for SPB and γ Dor stars computed by assuming that these pulsators are excited by convective flux blocking or the κ mechanism. This was already found in Paper I from *Gaia* DR3 data and also independently concluded from a summary of high-cadence *Kepler* and TESS space photometry by Balona (2024). We discuss the distributions of the properties of the pulsators, including the effect of the chosen threshold in the classification probability on our results.

The *Gaia* DR3 sample of stars having a probability above 50 per cent to be a g-mode pulsator from TESS is 23 times larger than the *Kepler* sample of 611 γ Dor stars that were asteroseismically characterised by Li et al. (2020). Starting from this *Kepler* sample, Fritzewski et al. (2024a) and Mombarg et al. (2024) determined the mass, convective core mass, radius, and age from grid-based modelling for subsets of respectively 490 and 539 of these pulsators, restricting to targets with identified prograde dipole modes and their measured asymptotic period spacing Π_0, also known as the buoyancy radius, or buoyancy travel time as it has the dimensions of a period. On the other hand, Li et al. (2020) estimated the internal rotation frequency near the convective core from the slope of the measured period spacing patterns from dipole prograde modes of consecutive radial order for all 611 genuine *Kepler* γ Dor stars with such identified modes. For the *Gaia* g-mode pulsators, we cannot perform asteroseismic modelling at this stage because their modes are not identified in terms of their geometry. This opportunity may occur in the future from their 5-year (or longer) yearly-interrupted TESS photometry. So far, such mode identification has only been achieved for stars in the Continuous Viewing Zones of TESS (Garcia et al. 2022b,a; Fritzewski et al. 2024b; Li et al. 2024).

Our aim for now is to prepare for future asteroseismic ensemble modelling of the *Gaia*-discovered TESS-confirmed g-mode pulsators by already deducing initial estimates for their masses, radii, convective core masses, and ages from the homogeneously deduced *Gaia* DR3 data. We do so for ∼14,000 new *Gaia*/TESS g-mode pulsators classified as such by Hey & Aerts (2024) with a probability above 50 per cent and having the same dominant frequency in both independent light curves. To reach our goal, we relied on their *Gaia* DR3 effective temperature and luminosity (obtained from the GSP-Phot analysis, Paper I), which are two observables available for all stars in the sample, thus allowing us to do a homogeneous sample analysis. To compensate for the generally underestimated uncertainties on the effective temperature, we inflated the uncertainties by a factor 3. This way, the typical uncertainty is more in line with what